Universidade Federal do Amazonas

Faculdade de Engenharia Elétrica e da Computação

Atividades - Disciplina Otimização

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1 Tópico 2

1.1

Consider the following function

$$f(x_1, x_2, x_3) = x_3 \log\left(e^{\frac{x_1}{x_3}} + e^{\frac{x_2}{x_3}}\right) + (x_3 - 2)^2 + e^{\frac{1}{x_1+x_2}}$$

Where the function $f : \mathbb{R}^3 \mapsto \mathbb{R}$ has its domain as
$\operatorname{dom} f = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_2 > 0, x_3 > 0\}$.
(a) Implement different numerical methods in Julia to optimize the given convex function
i. Gradient descent with backtracking line search. Use $t_{\text{init}} = 1, \alpha = 0.4, \beta = \frac{1}{2}$
ii. Newton's method (use same values for line search as part i)
Hints:

1. You can either calculate the gradient/Hessian analytically or use the finite difference method, i.e., $f'(x) = \frac{f(x+h)-f(x)}{h}$ where h is a small number like 10^{-5}.
2. Be cautious of the domain while implementing the numerical methods.

Take tolerance of 10^{-5} for the gradient norm wherever necessary.

1.2

Optimize the Problem 1 in Julia using one of the Quasi-Newton methods, namely BFGS (Broyden-Fletcher-Goldfarb-Shanno) method.

The below tutorial attempts to aid in understanding the method,
Quasi-Newton Methods: For the unconstrained, smooth convex optimization $\min_x f(x)$, where f is convex, twice differentable, and domain $= \mathbf{dom} f$. Earlier we had seen:

	Algorithm
Gradient-Descent	$x^+ = x - t\nabla f(x)$
Newton's Method	$x^+ = x - t\left(\nabla^2 f(x)\right)^{-1} \nabla f(x)$ or $x^+ = x - ts$ Where two main steps are to, Compute Hessian $\nabla^2 f(x)$ Solve the system $\nabla^2 f(x)s = -\nabla f(x)$
Now, Quasi-Newton	$x^+ = x - ts$ Where the direction s is defined by the linear system $Bs = -\nabla f(x)$ for some approximation B of $\nabla^2 f(x)$

With the information so far, we now state a Quasi-Newton template:
Let $x^{(0)} \in \mathbf{dom} f, B^{(0)} \succ 0$. For $k = 1, 2, 3, \ldots$, repeat:

1. Solve $B^{(k-1)} s^{(k-1)} = -\nabla f\left(x^{(k-1)}\right)$
2. Update $x^{(k)} = x^{(k-1)} + t_k s^{(k-1)}$
3. Compute $B^{(k)}$ from $B^{(k-1)}$

Final Note on Implementation:

- During the practical implementation of the algorithm in a compiler, the numerical value of $y_k^T s_k$ can get very small (since y_k involves difference of ∇f) which can lead to error of 1/0 form while calculating ρ_k. Thus, it can be a good idea to set ρ_k as a constant after $y_k^T s_k$ gets smaller than some certain ε say 10^{-5}.
- Take inital $H_0 = I$ (Identity matrix)

For more information, Refer:
Numerical Optimization (Jorge Nocedal and Stephen J. Wright), Springer, 2006. (Chapter 6)

1.3

3. Consider a well-stirred chemical reactor depicted in Figure 5, where an irreversible, firstorder reaction $A \to B$ occurs in the liquid phase. The feed to the reactor consists of pure A at flowrate F_0, temperature T_0, and molar concentration c_0. Note that the reactor temperature can be regulated with external cooling with temperature T_c.

Based on first principles, the following nonlinear state space model can be obtained which involves the mass and energy balances:

$$\begin{aligned}
\frac{dc}{dt} &= F_0\left(c_0 - c\right) - k_0 \exp\left(-\frac{E}{RT}\right) c \\
\frac{dT}{dt} &= \frac{F_0\left(T_0 - T\right) - \Delta H}{\pi r^2 h} + \frac{k_0 \exp\left(-\frac{E}{RT}\right) c + \frac{2U}{r\rho c_p}\left(T_c - T\right)}{\pi r^2} \\
\frac{dh}{dt} &= \frac{F_0 - F}{\pi r^2}
\end{aligned}$$

The controlled variables are the level of the tank, h, and the molar concentration of species A , denoted as c; the additional state variable is the reactor temperature, denoted as T, While, the manipulated variables are the coolant liquid temperature, T_c, and the outlet flowrate, F. The open-loop stable steady-state operating conditions can be obtained with specified model parameters and are the following

$$\begin{aligned}
c^s &= 0.878\text{kmol/m}^3 \qquad & T^s &= 324.5\ \text{K} h^s = 0.659\ \text{m} \\
T_c^s &= 300\ \text{K} & F^s &= 0.1\ \text{m}^3/\text{min}
\end{aligned}$$

Using a sampling time of 1 min , a linearized discrete state space model is obtained to approximate the complex process model (i.e., the set of ODE equations). It is also assumed that all the states are measurable without noise. Shifted by the given steady state, the state space variables are

$$x = \begin{bmatrix} c - c^s \\ T - T^s \\ h - h^s \end{bmatrix} \quad u = \begin{bmatrix} T_c - T_c^s \\ F - F^s \end{bmatrix} \quad y = \begin{bmatrix} c - c^s \\ T - T^s \\ h - h^s \end{bmatrix}$$

The corresponding linear model is

$$\begin{aligned}
x(k+1) &= Ax(k) + Bu(k) \\
y(k) &= Cx(k)
\end{aligned}$$

in which

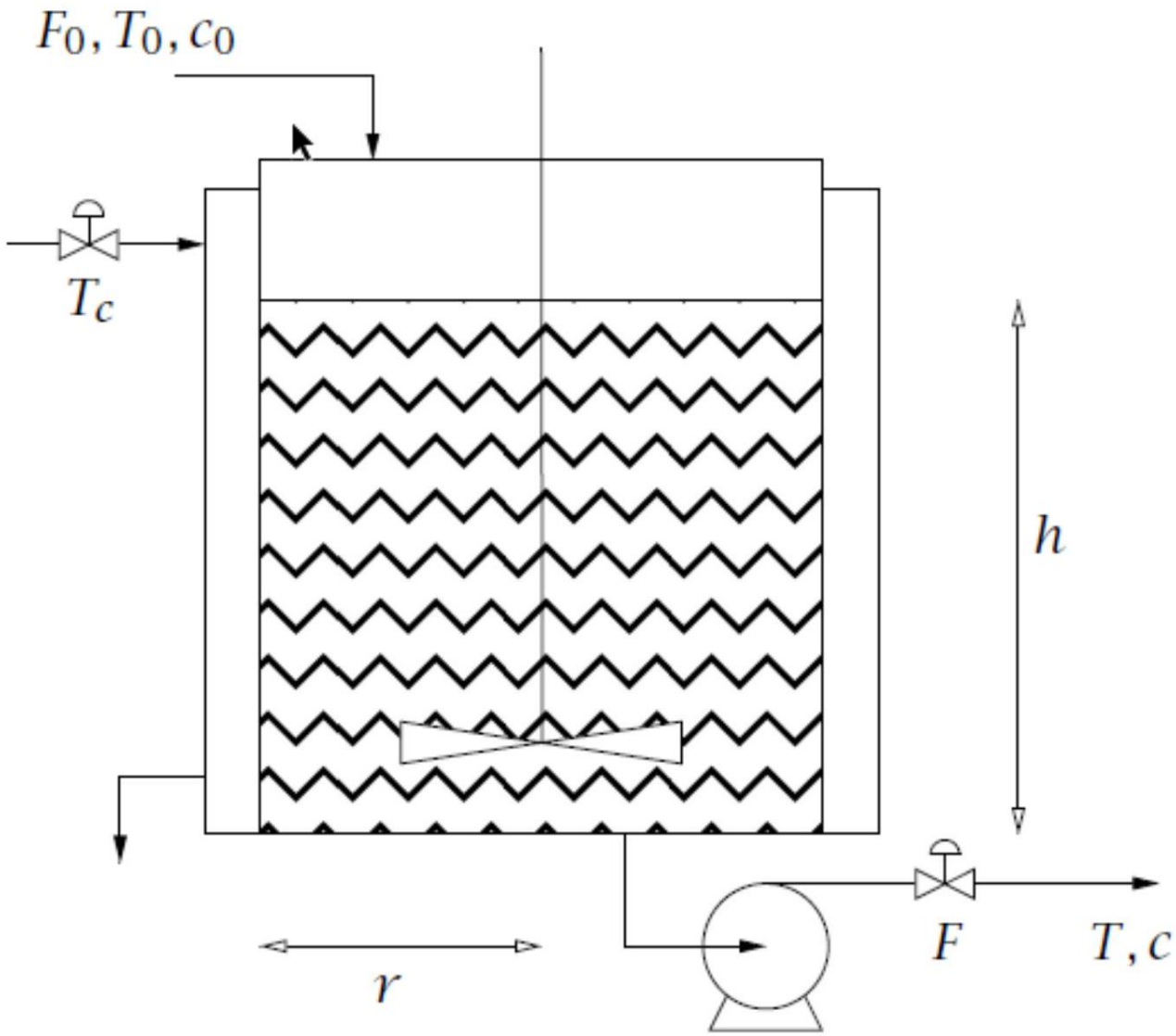

Figure 1: Schematic of the continuous stirred tank reactor.

$$A = \begin{bmatrix} 0.2681 & -0.00338 & -0.00728 \\ 9.703 & 0.3279 & -25.44 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} -0.00537 & 0.1655 \\ 1.297 & 97.91 \\ 0 & -6.637 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The goal in this case is to develop an optimization problem to drive the two controlled variables, c and h from the given initial value to the specified constant setpoints. Specifically, the considered time instants are from 0 to 3 (i.e., $k = 0, 1, 2, 3$), and the initial states and the setpoints for the two controlled variables are given as

$$x(k=0) = \begin{bmatrix} -0.03 \\ 0 \\ 0.3 \end{bmatrix} \quad c_{sp} = 0 \quad h_{sp} = 0$$

In this sense, the objective function is to minimize the deviation of the two controlled variables from their setpoints over the time horizon, which can be derived as follows

$$\min \sum_k \left(|y_1(k) - c_{sp}| + |y_3(k) - h_{sp}| \right)$$

Also, the bounds for the input and state variables are given as follows

$$\begin{bmatrix} -0.05 \\ -5 \\ -0.5 \end{bmatrix} \leq x(k) \leq \begin{bmatrix} 0.05 \\ 5 \\ 0.5 \end{bmatrix} \quad \begin{bmatrix} -10 \\ -0.05 \end{bmatrix} \leq u(k) \leq \begin{bmatrix} 10 \\ 0.05 \end{bmatrix}$$

Derive the LP problem for this purpose with the given information of the state space model, time horizon, initial state, setpoints, and bounds. Then, use Julia/JuMP to solve the problem.